

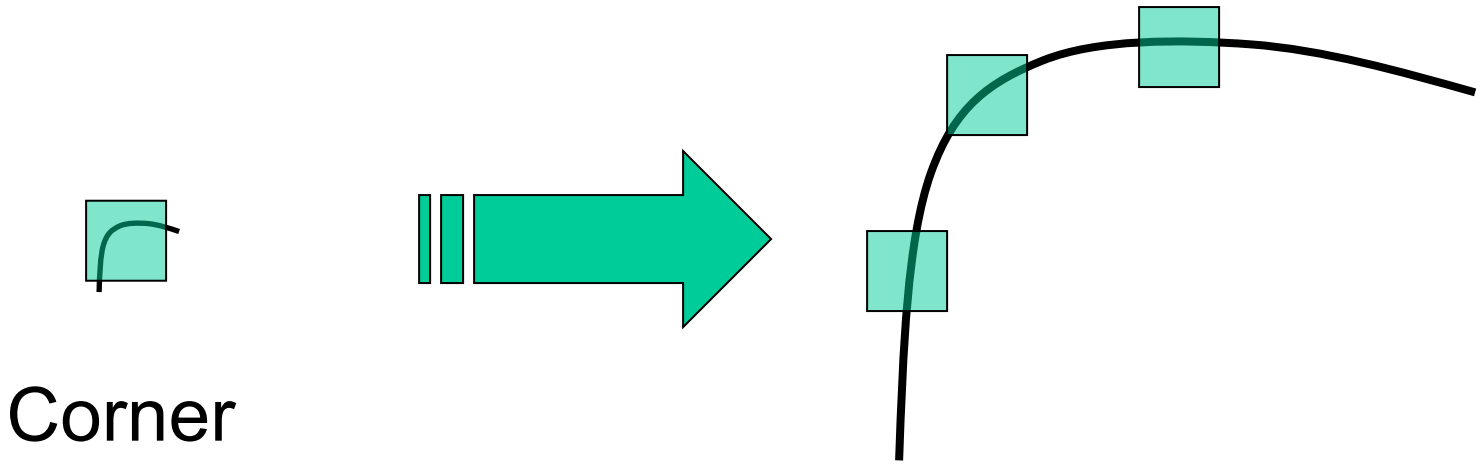
上海科技大学
ShanghaiTech University

SIFT Feature

CS172 Computer Vision I

Instructor: Jiayuan Gu

Harris Corner Detection is not Scale-Invariant



Corner

All points will be classified as edges

Corner location is not invariant w.r.t. scaling

Source: L. Lazebnik

Multi-Scale Representation: Image Pyramid

One solution is to extract features at a variety of scales

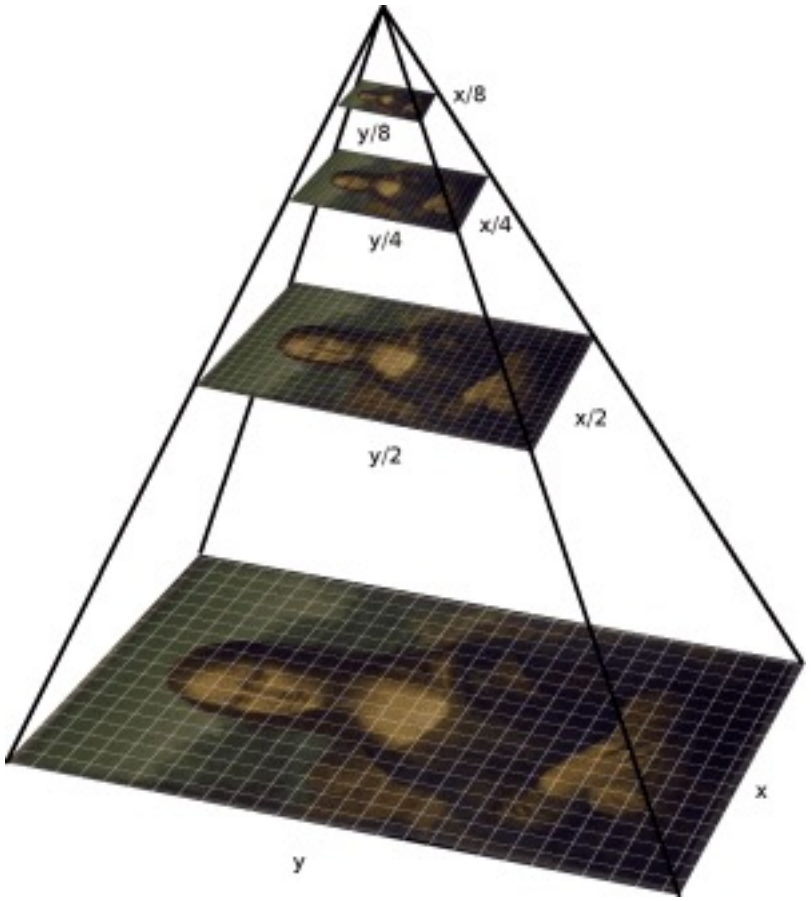
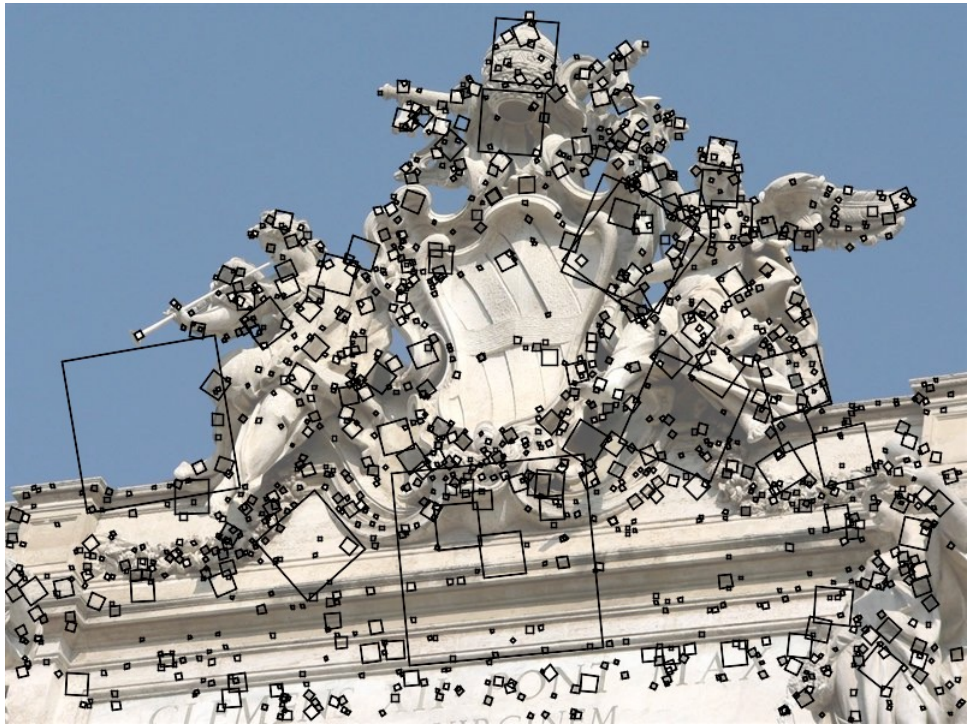


image source

SIFT Features

Detected features with characteristic scales and orientations





David G. Lowe. **"Distinctive image features from scale-invariant keypoints."** *IJCV* 60 (2), pp. 91-110, 2004.

Outline

- Roadmap
  - Feature detection
    - Select the characteristic scale
    - Localize the keypoint
  - Feature description
    - Assign the orientation
    - Describe the keypoint

- Learning Objectives
  - Explain and derive scale-normalized Laplacian of Gaussian
  - Understand the image pyramid
  - Construct and explain a SIFT descriptor

Feature Detection

Localize multi-scale distinguishing features

Basic Idea: Detect Blobs

Convolve the image with a “blob filter” at multiple scales and look for extrema of filter response in the resulting scale space

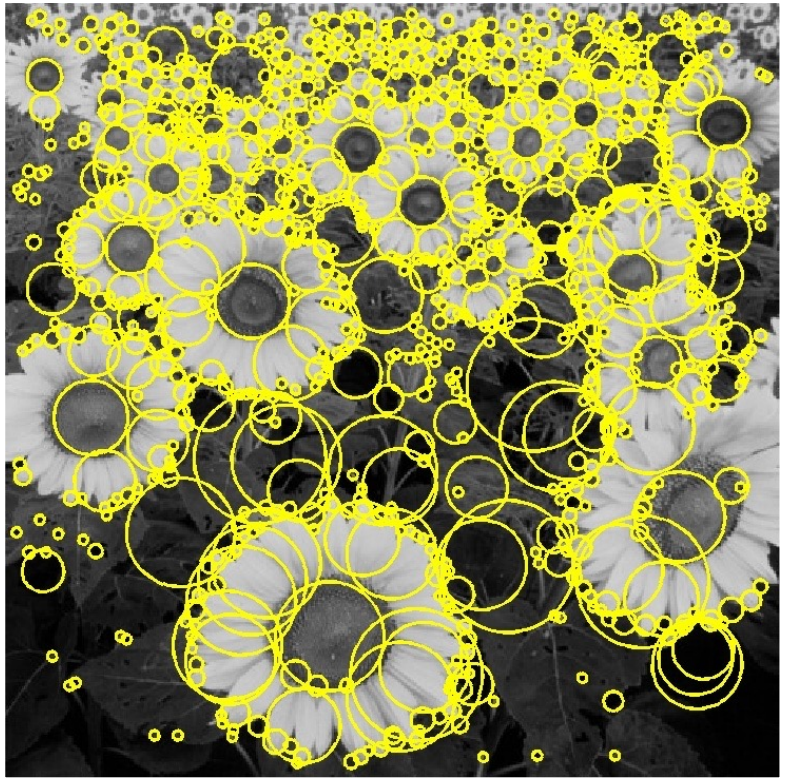
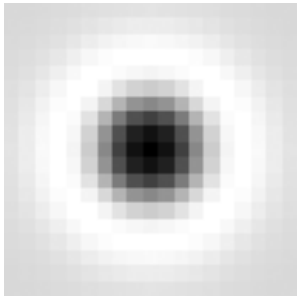


image with detected blobs



filter

T. Lindeberg, Feature detection with automatic scale selection, *IJCV* 30(2), pp 77-116, 1998

Source: L. Lazebnik

Detect Blobs with Convolution



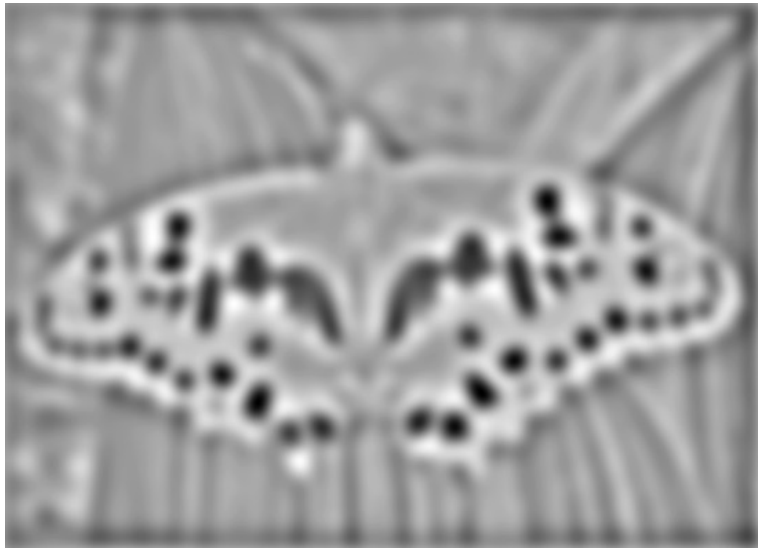
Input image

*



Filter/kernel

=

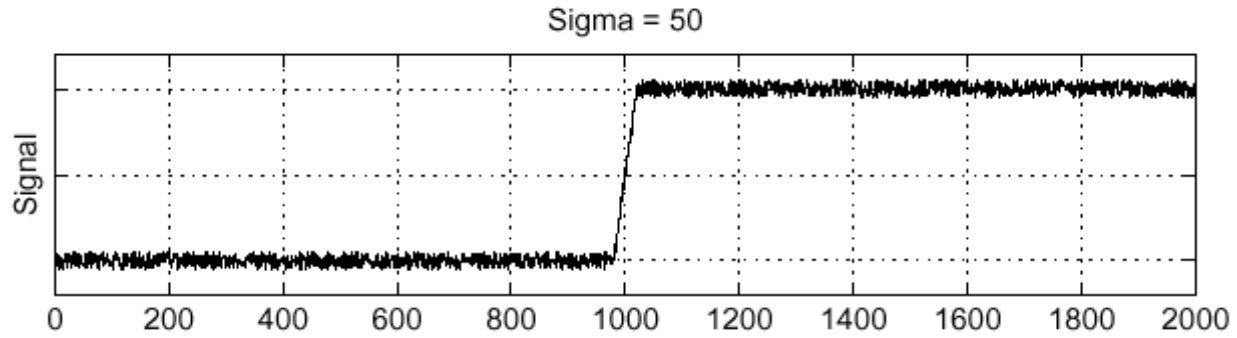


Convolved result

Source: N. Snavely

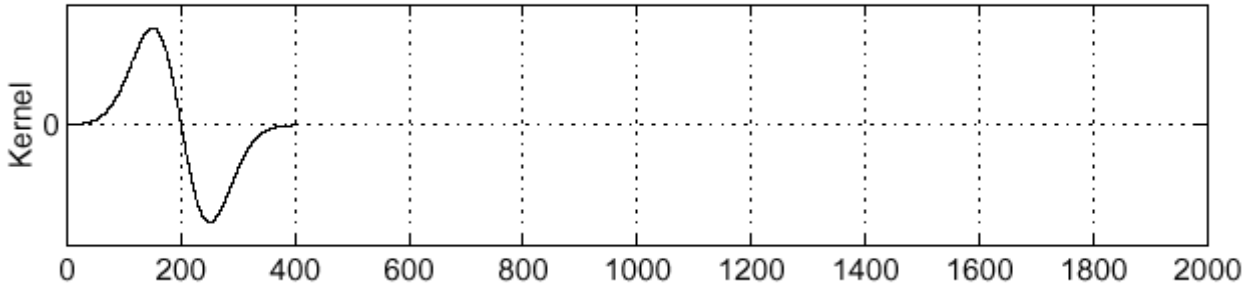
How to design a blob filter?

f



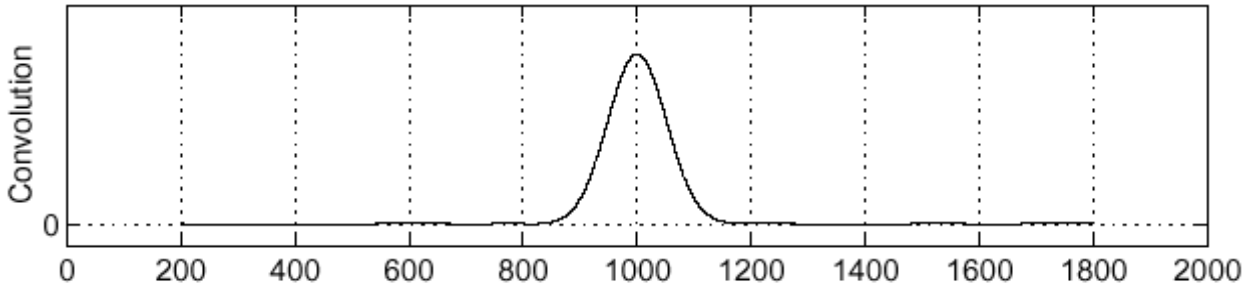
Signal

$\frac{d}{dx} g$



Derivative of Gaussian

$f * \frac{d}{dx} g$



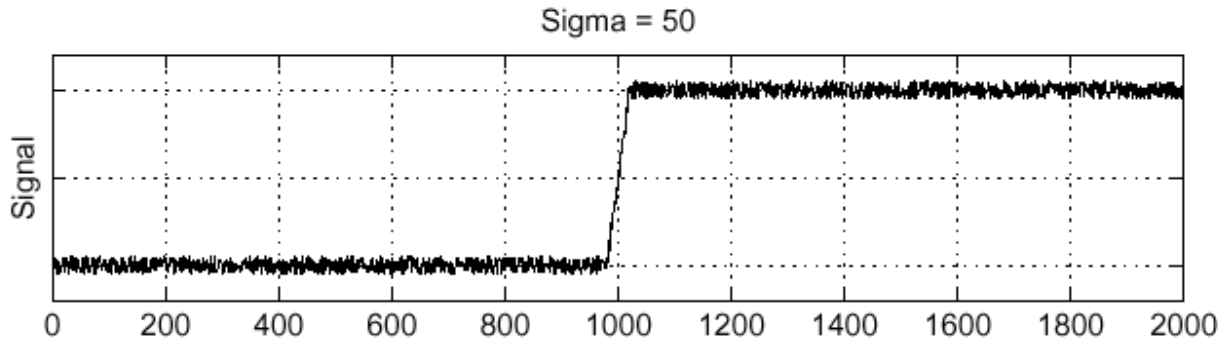
Edge is detected at the maximum of derivative

Recall edge detection

Source: S. Seitz

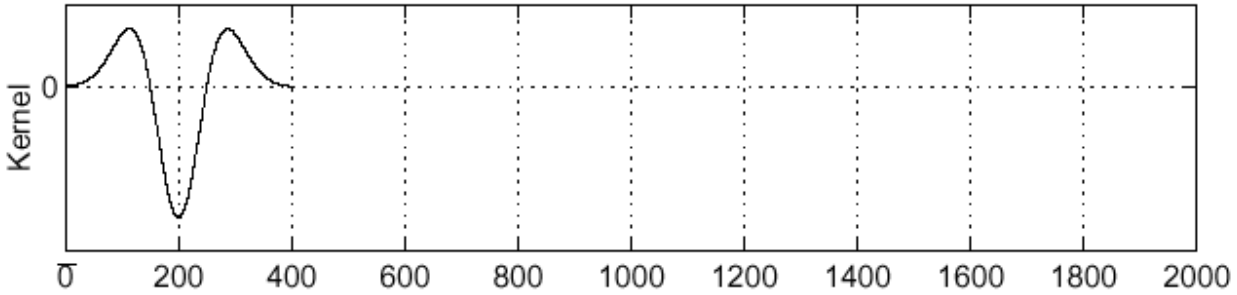
Where is the edge?

f



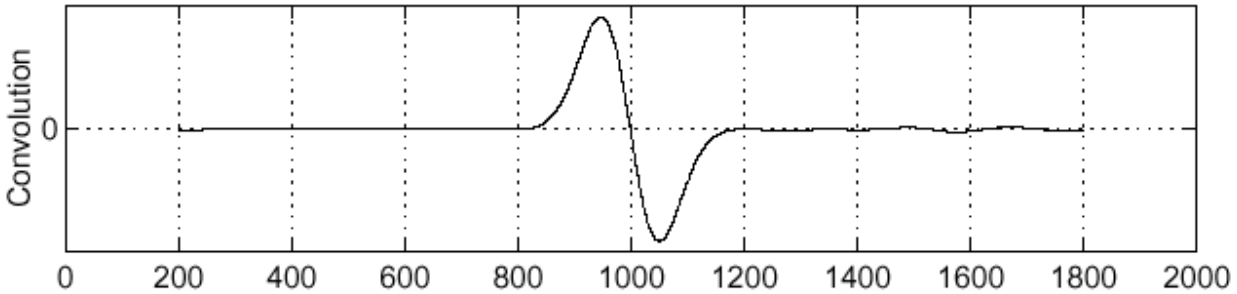
Signal

$\frac{d^2}{dx^2} g$



Second derivative of Gaussian (Laplacian)

$f * \frac{d^2}{dx^2} g$

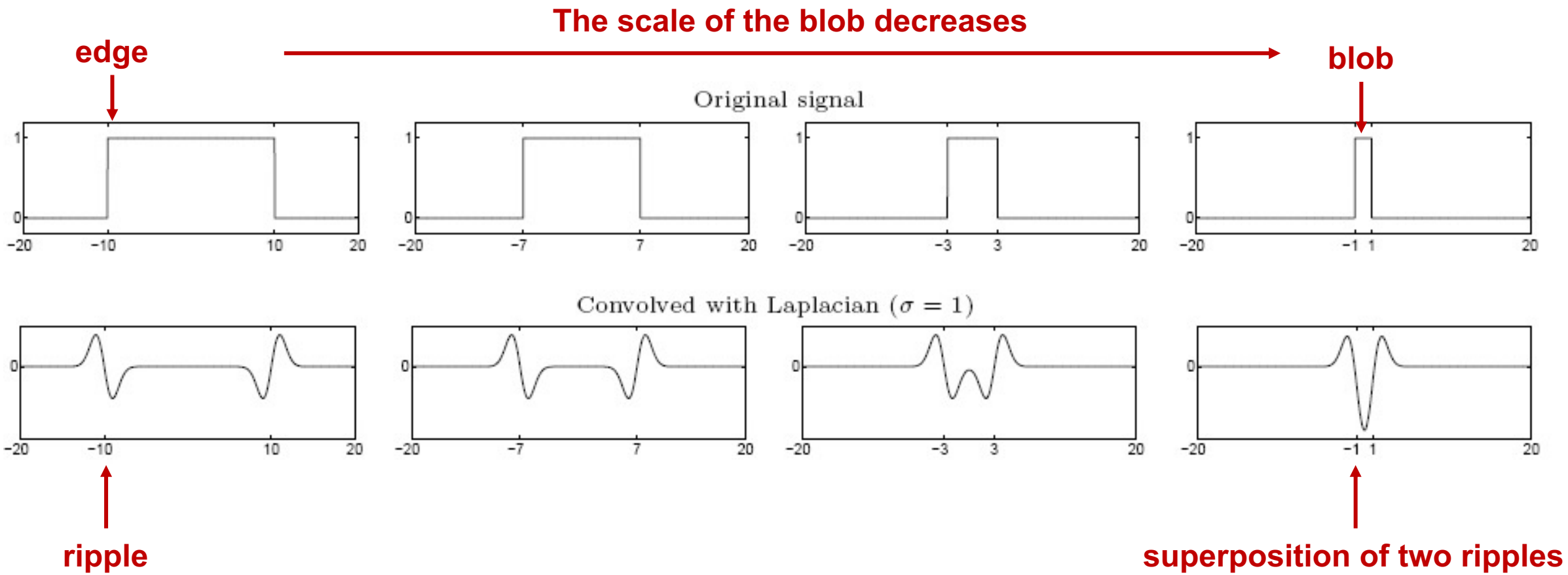


Edge is zero crossing of second derivative

Take the second derivative

Source: S. Seitz

From Edge to Blob

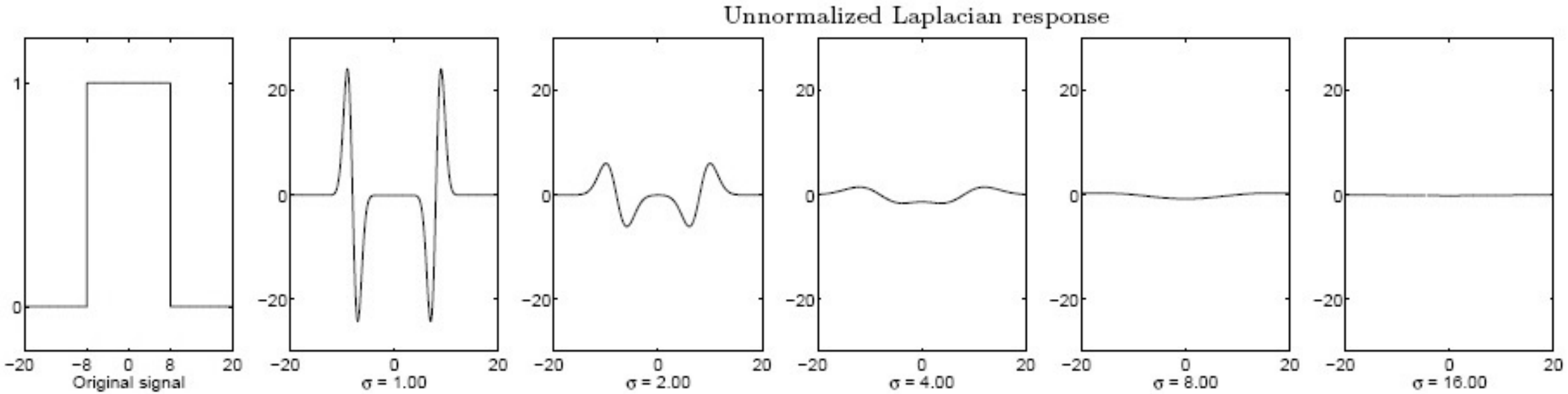


Spatial: the magnitude of the Laplacian response will achieve a maximum at the center of the blob

Scale: the scale of the Laplacian is “matched” to the scale of the blob

Credit: L. Lazebnik

From Edge to Blob



original signal
(radius=8)

Increasing the scale σ

Laplacian response decays as scale increases

Credit: L. Lazebnik

Why is normalization needed?

- When the Gaussian scale σ is fixed and the blob scale decreases, the center response magnitude first increases and then decreases

- When the blob scale is fixed and the Gaussian scale σ increases, the center response magnitude decays

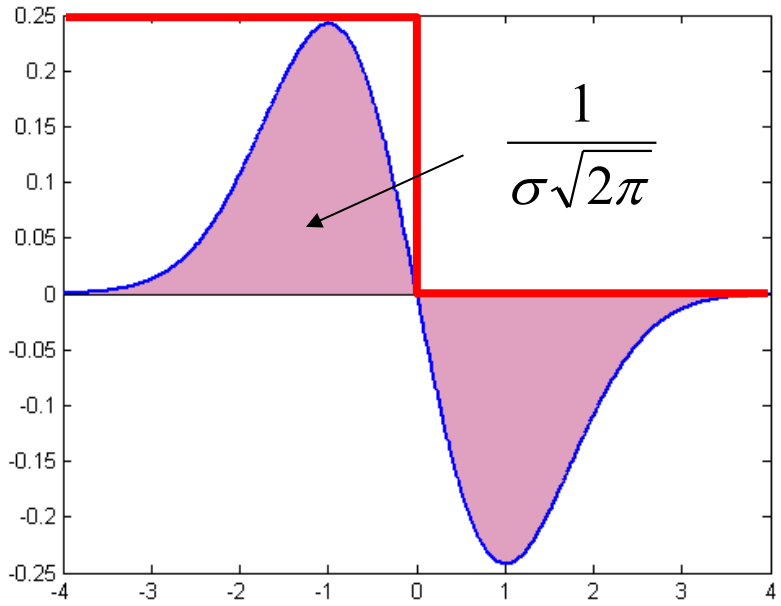
- We want to find the characteristic scale of the blob by convolving it with Laplacians at several scales and looking for the maximum response

Scale Normalization

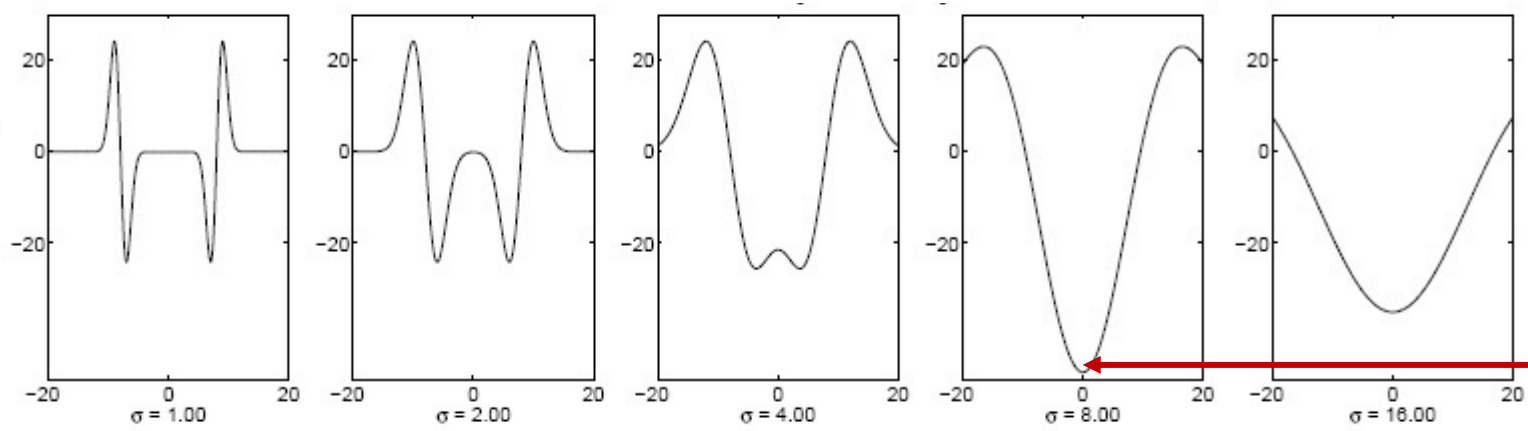
The response of a derivative of **Gaussian filter** to a perfect step edge decreases as σ increases

To fairly compare and find the actual maximum response, we want to make the max **Laplacian** response scale-invariant

Laplacian is the second Gaussian derivative, and thus normalized by σ^2



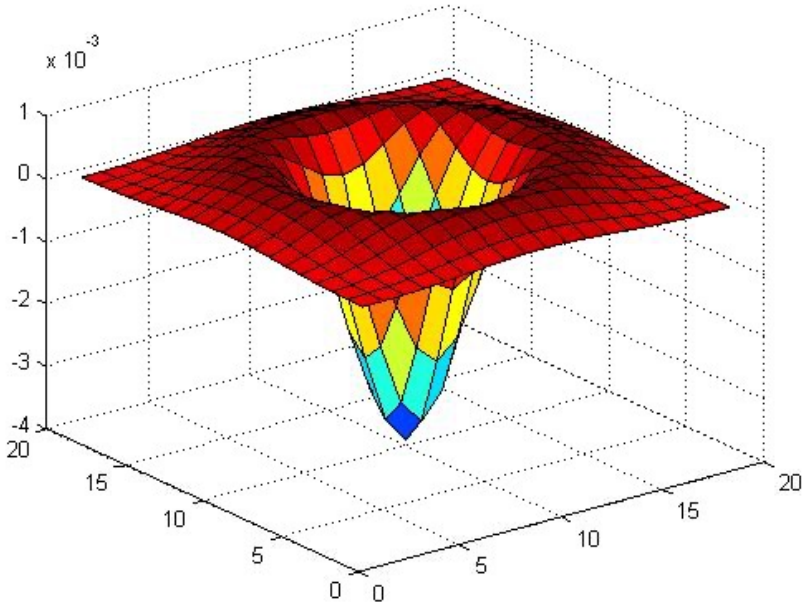
Scale-normalized Laplacian response

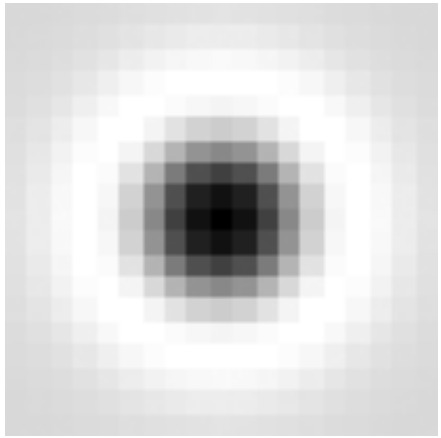


maximum response at characteristic scale

Laplacian of 2D Gaussian (LoG)

Laplacian of Gaussian: Circularly symmetric operator for blob detection in 2D





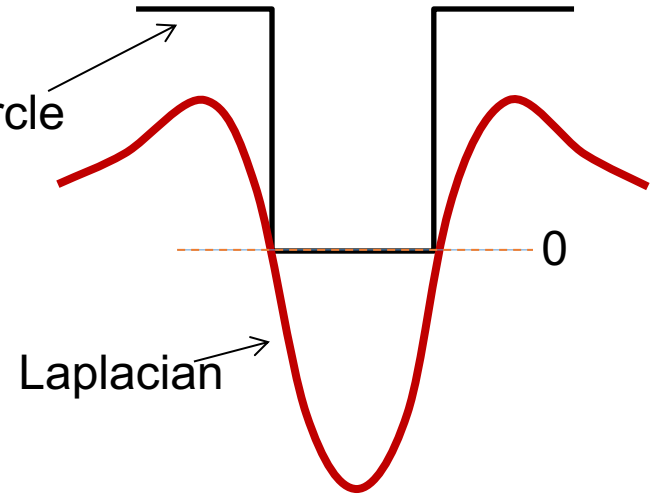
$$\nabla^2 g = \frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial y^2} \xrightarrow{\text{scale normalization}} \nabla^2_{\text{norm}} g = \sigma^2 \left(\frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial y^2} \right)$$

Credit: L. Lazebnik

2D Blob Detection

- At what scale, does the Laplacian achieve a maximum response to a binary circle of radius r?
- To get the maximum response, the zeros of the Laplacian must be aligned with the circle
- The Laplacian of Gaussian is computed as:

$$-\frac{1}{\pi\sigma^4}\left(1-\frac{x^2+y^2}{2\sigma^2}\right)e^{-(x^2+y^2)/2\sigma^2}$$



- The maximum response is achieved at $r^2 = 2\sigma^2$

Credit: L. Lazebnik

Scale-Space Blob Detector

1. Convolve the image with scale-normalized Laplacian at several scales

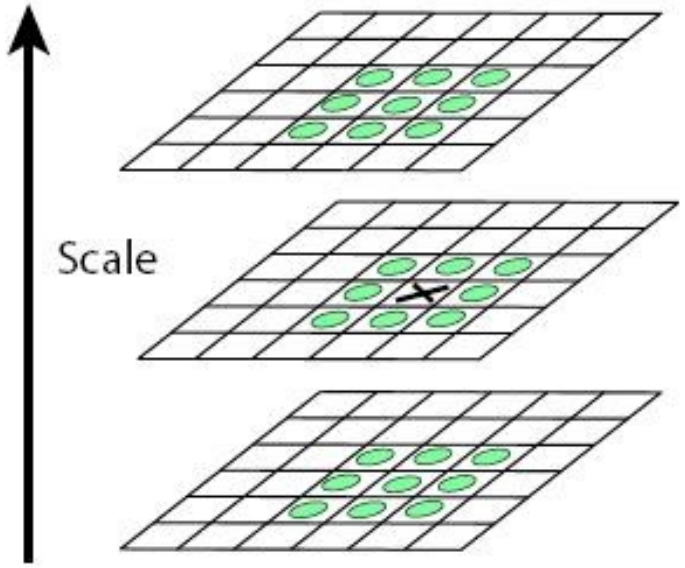


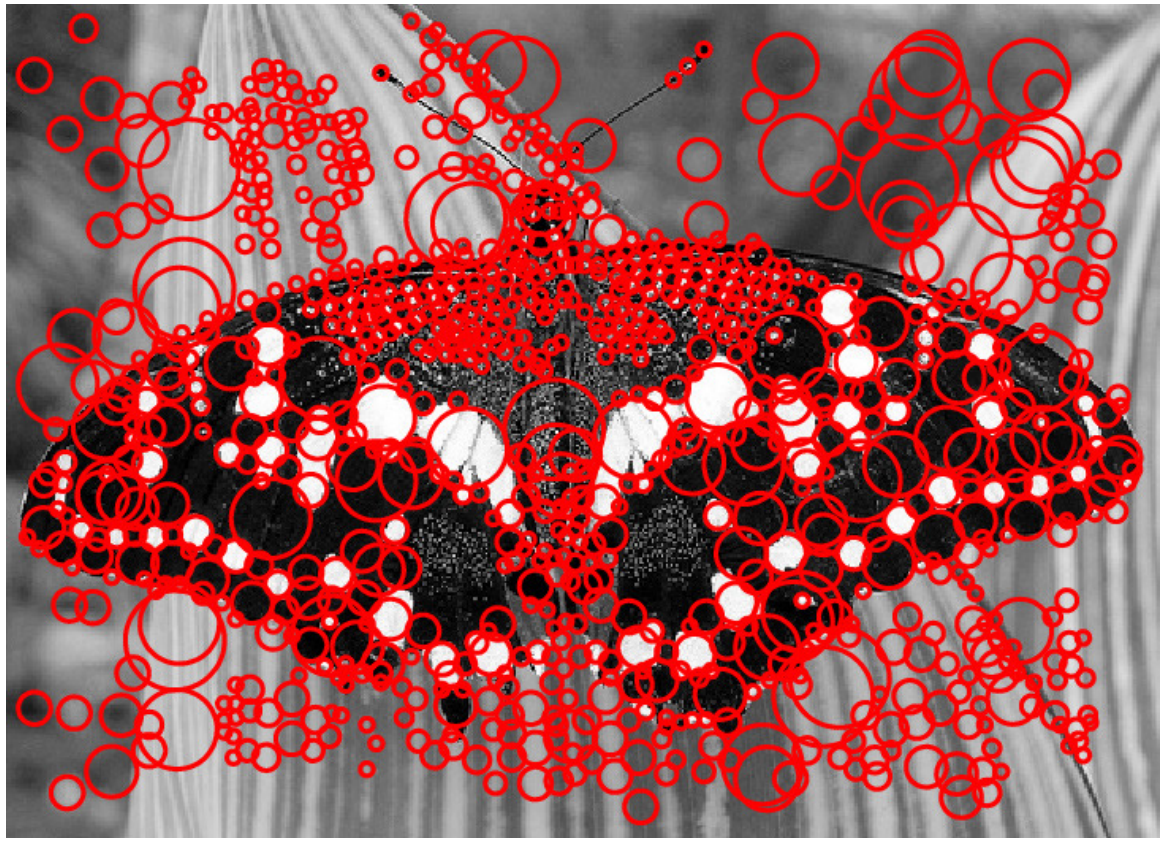


sigma = 11.9912

Scale-Space Blob Detector

1. Convolve the image with scale-normalized Laplacian at several scales
2. Find the maxima of Laplacian response in the scale space (3D maximum)





Efficient Implementation of LoG

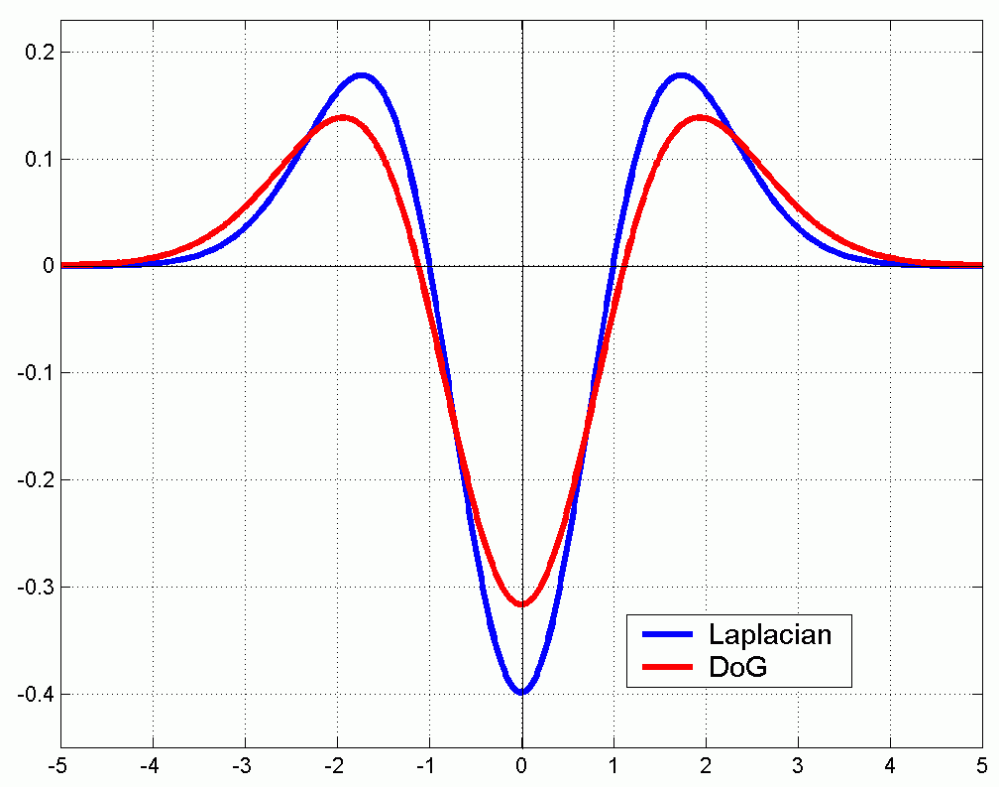
Approximating the Laplacian with a difference of Gaussians (DoG)

$$L = \sigma^2 \left(G_{xx}(x, y, \sigma) + G_{yy}(x, y, \sigma) \right)$$

(Laplacian)

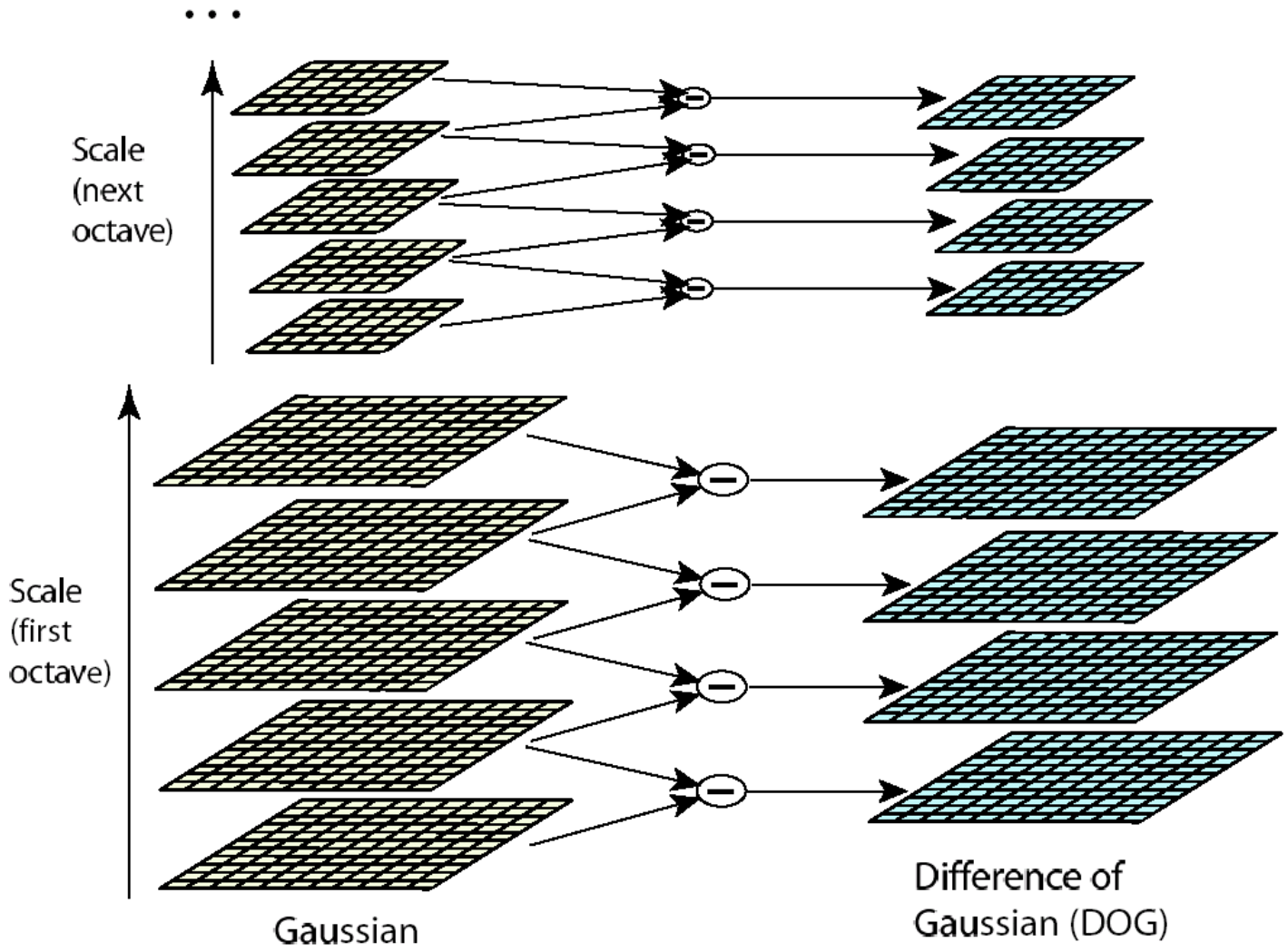
$$DoG = G(x, y, k\sigma) - G(x, y, \sigma)$$

(Difference of Gaussians)



Source: L. Lazebnik

From Gaussian Pyramid to DoG Pyramid



David G. Lowe. **"Distinctive image features from scale-invariant keypoints."** *IJCV* 60 (2), pp. 91-110, 2004.

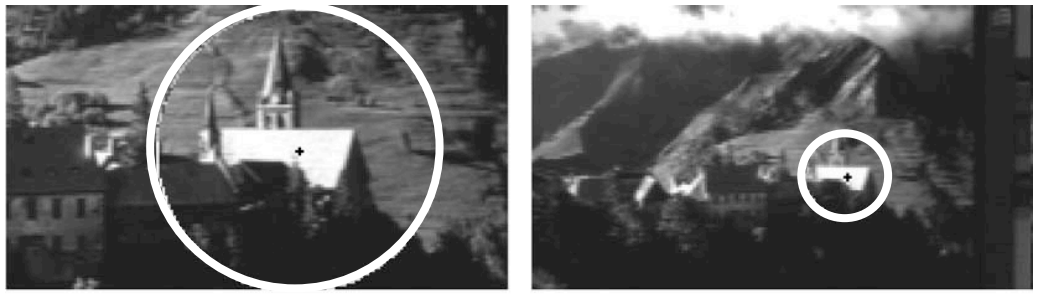
Feature Descriptor

Describe the features of detected keypoints

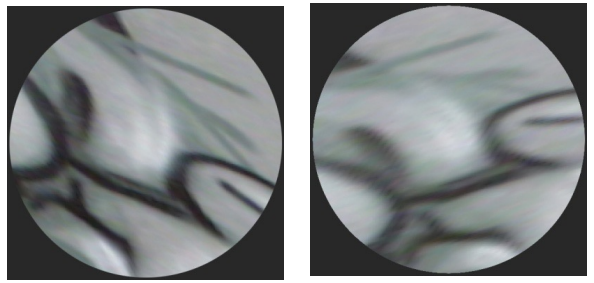
From Feature Detection to Feature Description

- Even if the detector finds both patches correctly, their raw pixel arrays are different

- To recognize a pattern, we need to describe it with a ***signature*** of the neighborhoods of extracted keypoints
  - Such signatures are expected to be transformation-invariant



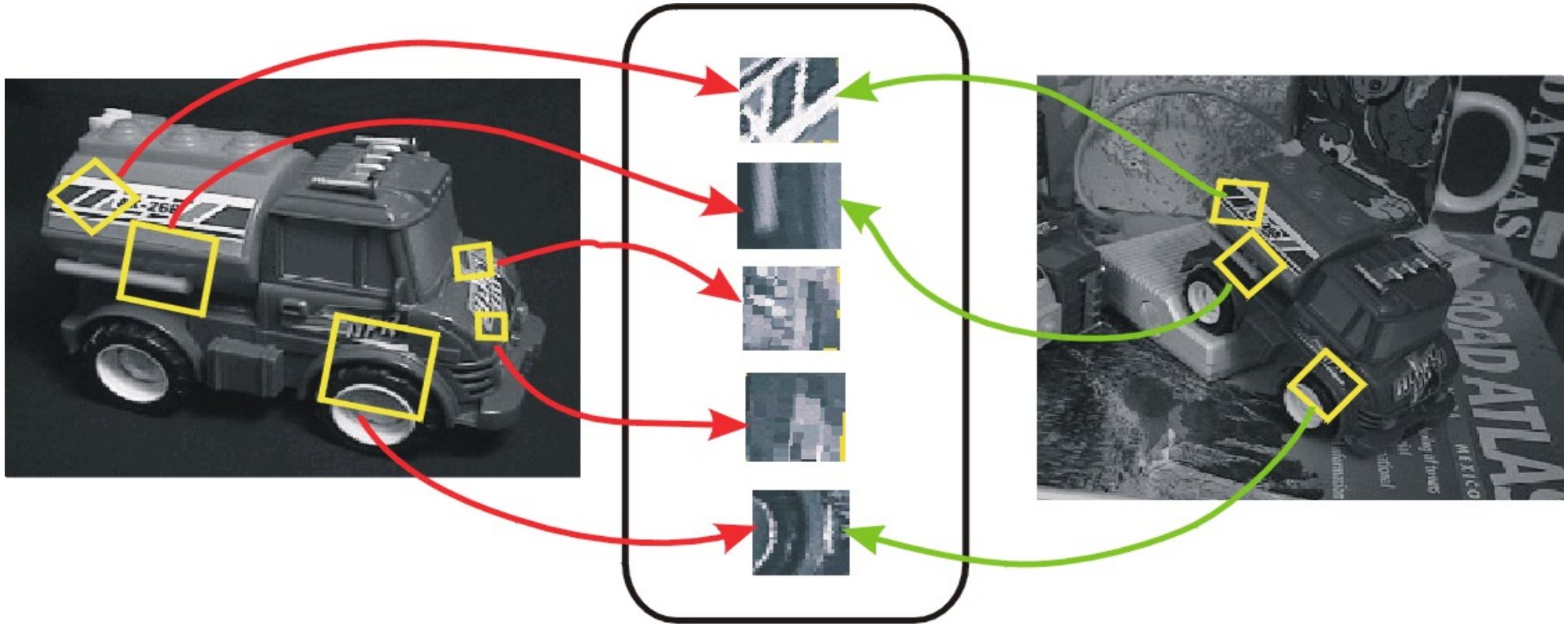
scale change



rotation change

Source: L. Lazebnik

From Keypoint Detection to Feature Description

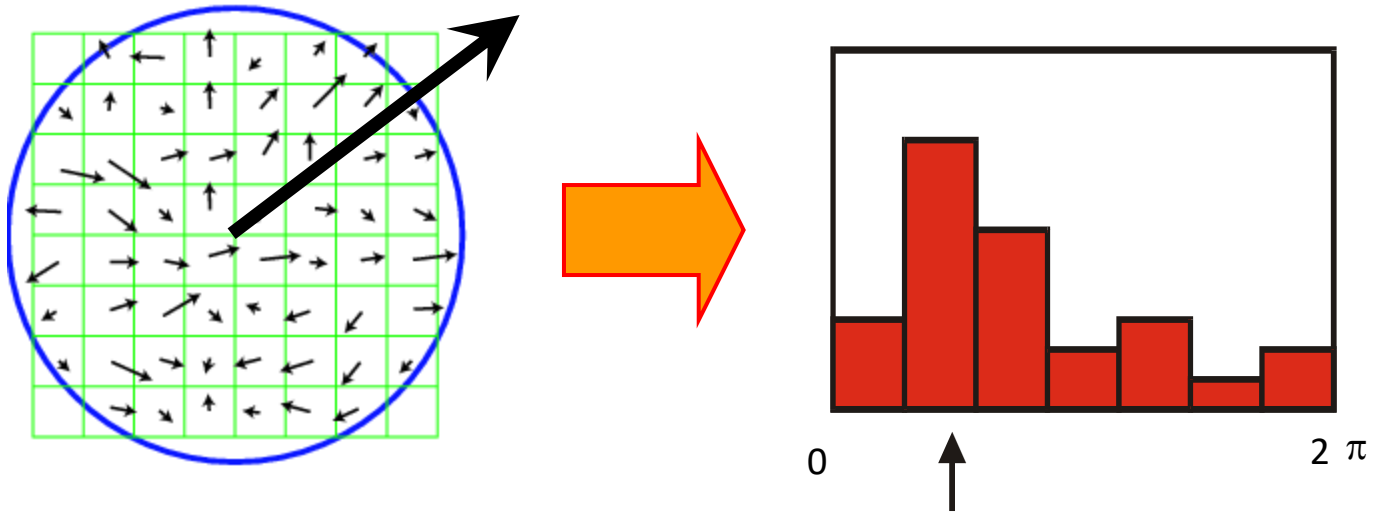


Detection D is ***covariant***: $D\big(T(I)\big) = T\big(D(I)\big)$
Description F is ***invariant***: $F\big(T(I), T(p)\big) = F(I, p)$
where I is the image and T is the transformation

Source: L. Lazebnik

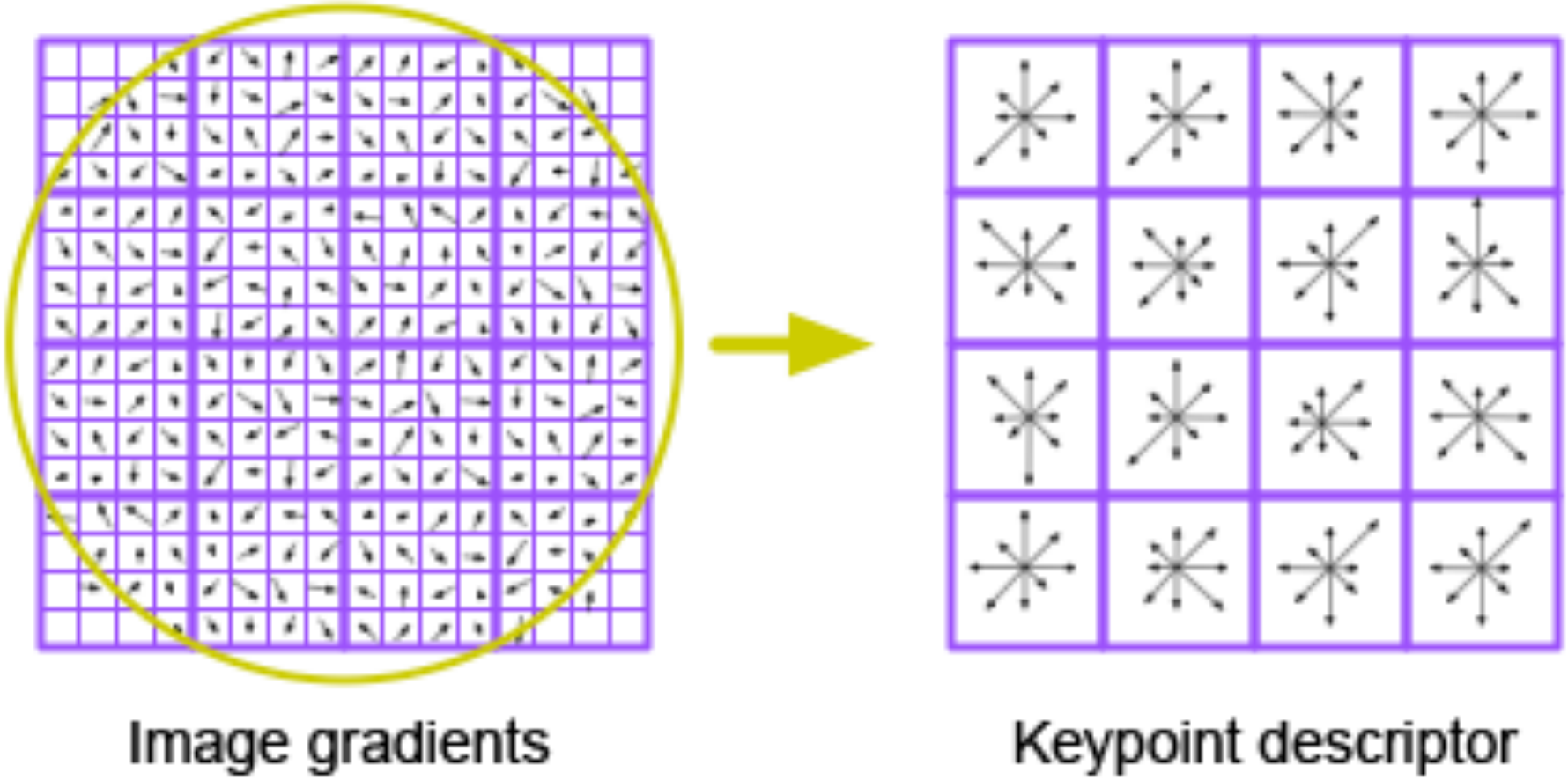
Key Idea: Assign a Reference Orientation

- At the selected scale
  - Compute local gradient directions
  - Create the histogram of local gradient directions within the patch
  - Assign the reference orientation at the peak of the smoothed histogram
- Normalize the local gradient directions given the reference orientation, to achieve rotation invariance



Source: L. Lazebnik

SIFT Descriptor (128-dim)



Divide the canonical **16x16 patch** into a **4×4 spatial grid (pooling)**
Within each cell, record an **8-bin gradient orientation histogram**

What SIFT has normalized away

Variation	Mechanism
Translation	detect local keypoint
Scale	select characteristic scale and scale patch
In-plane rotation	dominant orientation
Contrast scaling / Illumination	descriptor normalization
Small deformation / localization errors	spatial/orientation pooling





SIFT Properties

- Extraordinarily robust detection and description technique
- Can handle changes in viewpoint
  - Up to about 60 degree out-of-plane rotation
- Can handle significant changes in illumination
- Fast and efficient—can run in real time

Summary

- SIFT’s pipeline
  - Detection and select scale: extrema in space + scale
  - Orient: establish a canonical frame
  - Describe: pool local gradient structure into $4 \times 4 \times 8 = 128$ dimensions

Reading List

- **Szeliski, *Computer Vision: Algorithms and Applications*, 2e.**
  - Ch. 7.1.1–7.1.3: Feature Detectors, Descriptors, and Matching.
- **Forsyth & Ponce, *Computer Vision: A Modern Approach*, 2e.**
  - Ch. 5.3–5.5; review Ch. 4.7 for Gaussian pyramids.
- **Lowe, IJCV 2004, "Distinctive Image Features from Scale-Invariant Keypoints."**
- **Practical:** OpenCV, "Introduction to SIFT."
- **Optional advanced:** Lindeberg, IJCV 1998, "Feature Detection with Automatic Scale Selection."